# LOMBA KOMPETENSI SISWA PENDIDIKAN MENENGAH

**TINGKAT PROVINSI DKI JAKARTA**

**TAHUN 2025**


# TEST PROJECT

# WINDOWS ENVIRONMENT

## IT NETWORK SYSTEMS ADMINISTRATION

# Introduction

You are working as a system administrator for the itnsa.id organization. This organization utilizes products from Microsoft for providing resources and services. Meanwhile, Ansible is used as the open-source solutions for automation purposes. As a good system administrator, you must be able to complete all of these tasks within fixed time.

Main purpose of this test project is to provide services and resources like active directory for centralized management, web service, file sharing, and automation using ansible. **In addition**, you must ensure all hosts inside itnsa.id can access the INET server. You don't need to configure anything on INET, just leave it as it is.

**Credential Information**
**Windows**
Username: User /
Administrator Password:
P@ssw0rd2025

**Linux**
Username: root / user
Password: P@ssw0rd2025

# Description of project and tasks

## Basic Configuration

Configure hostname, FQDN and IP address on all hosts of itnsa.id refer to the information table and set timezone to **(UTC+07:00) Bangkok, Hanoi, Jakarta**.

## Active Directory

**dc.itnsa.id**

1. Configure initial domain controller (new forest) for **itnsa.id** domain
2. Join **srv, fw,** and **workstation** host to **itnsa.id** domain
3. Create the following organizational units (OU) inside **itnsa.id** domain:
   a) Direksi
   b) Manager
   c) Karyawan
4. Create following AD groups inside their corresponding OU:
   a) Direksi
   b) Manager
   c) Karyawan
5. Create AD users refer to the table below and make sure user doesn't need to change their password at logon:

| Username | Password | Group | OU |
|---|---|---|---|
| drs1-10 | P@ssw0rd | Direksi | Direksi |
| mng1-50 | P@ssw0rd | Manager | Manager |
| kyw1-100 | P@ssw0rd | Karyawan | Karyawan |

   *Note:*

   *drs1-3 means you must create drs1, drs2 and drs3. This applied to the other like **mng1, mng2** until **mng10** same as **kyw1, kyw2, kyw3** until **kyw100***

6. All created domain users must use **\\filesrv.itnsa.id\Home\%username%** as their home drive and it must be mapped into **H:** drive.
7. Configure group policy and name it as **MAIN GPO** to the prevent the welcome animation from appearing on first login

## DNS Service

**dc.itnsa.id**

1. Create DNS record refer to the table below:

| Type | Record | Value |
|---|---|---|
| NS | @ | dc.itnsa.id |
| A | dc | 172.16.0.1 |
| A | srv | 172.16.0.10 |
| A | fw | 172.16.0.254 |
| CNAME | www | srv.itnsa.id |
| CNAME | private | srv.itnsa.id |
| PTR | 172.16.0.1 | dc.itnsa.id |
| PTR | 172.16.0.10 | srv.itnsa.id |
| PTR | 172.16.0.254 | fw.itnsa.id |

2. Set DNS root hint **ONLY** to the IP address of INET server.
3. Set DNS forwarder **ONLY** to the IP address of INET server.

## Certificate Authority

**dc.itnsa.id**

**1.** Configure Enterprise Root CA for itnsa.id. Set common name to **ITNSA-CA**

**2.** Distribute Root CA certificate to all host inside itnsa.id site, excluding **ansible-srv**

3. Use certificate generated by this CA for securing any services that needs certificate

## DHCP Service

**fw.itnsa.id**

1. Configure DHCP for **itnsa.id** clients with specification refer to the list below:
    - Scope name: Internal
    - Network: 172.16.0.0/24
    - Range: 172.16.0.100 – 172.16.0.250
    - Exclude: 172.16.0.200 – 172.16.0.210
    - Default gateway: 172.16.0.254
    - Default DNS: 172.16.0.1
    - Default Domain: itnsa.id
    - Duration: 1 days, 1 hours, 1 minutes
2. Make sure DHCP client can automatically register their hostname into domain zone of itnsa.id

## RAID

**srv.itnsa.id**

1. Add 3 extra disks with each size **10GB**
2. Format the RAID volume using **NTFS** filesystem
3. Assign formatted RAID volume into **R:** drive

## File Service

**srv.itnsa.id**

1. Configure shared folder refer to the table below:

| Share Name | Directory | Permission |
|---|---|---|
| Direksi | R:\Files\Direksi | Direksi |
| Manager | R:\Files\Manager | Manager |
| Karyawan | R:\Files\Karyawan | Karyawan |
| Home | R:\Files\Home | Domain Users |

2. Set quota to 100MB for each group shared folder and 50MB for home shared folder.
3. Prevent executable files like **.bat** and **.ps1** from being saved on group and home shared folder.

## IIS Web Service

**srv.itnsa.id**

1. Each created website below must be secured with certificate signed by **ITNSA-CA**
2. Configure **www.itnsa.id** website with specification below:
    - Set website root directory into **R:\inetpub\www**
    - Set website content to "**Welcome to www.itnsa.id**"
3. Configure **private.itnsa.id** website with specification below:
    - Set website root directory into **R:\inetpub\private**
    - Set website content to "**Private site of itnsa.id**"
    - Only **Direksi** and **Manager** group can access this website

## Routing

**fw.itnsa.id**

1. Install Routing and Remote Access feature
2. Configure Routing feature to make fw can forward traffic from internal client of itnsa.id
3. Configure NAT to make traffic from itnsa.id to INET translated into public IP address of fw
4. After Routing & NAT configured, make sure workstation can access **www.public.net**

## INET

Configure INET as the authoritative DNS server of public.net and also host www.public.net website.

# Ansible

**ansible-srv**

IP address**,** Ansible inventory, and other utility for automation inside must be done only on **ansible-srv**. Ansible inventory should be placed on **/etc/ansible/hosts**. You need to create the following playbooks:

1. Create playbook **/etc/ansible/install_features.yml** to install NFS-Client and TFTP-Client features on all hosts defined in the ansible inventory
2. Create playbook **/etc/ansible/dns_record.yml** to create a DNS record with type A on **itnsa.id** domain zone. Use `record` and `value` as the variables for the playbook.

For example, creating **test.itnsa.id** record with value **172.16.0.1** can be done using this command:

```
# ansible-playbook /etc/ansible/dns_record.yml -e record=test -e value=172.16.0.1

Playbook executed...
```

3. Create playbook **/etc/ansible/shared_folder.yml** to create a shared folder on SRV. Use list of variables below for the playbook:
   - `share_name` for shared folder name.
   - `path` for shared folder path.
   - `access` for specify active directory user or group that has **Read** permission on shared folder.
   - `owner` for specify active directory user or group that has **Full** permission on shared folder. This variable is optional, so if it's not defined, the created shared folder should NOT have any Full permission given to active directory user or group.

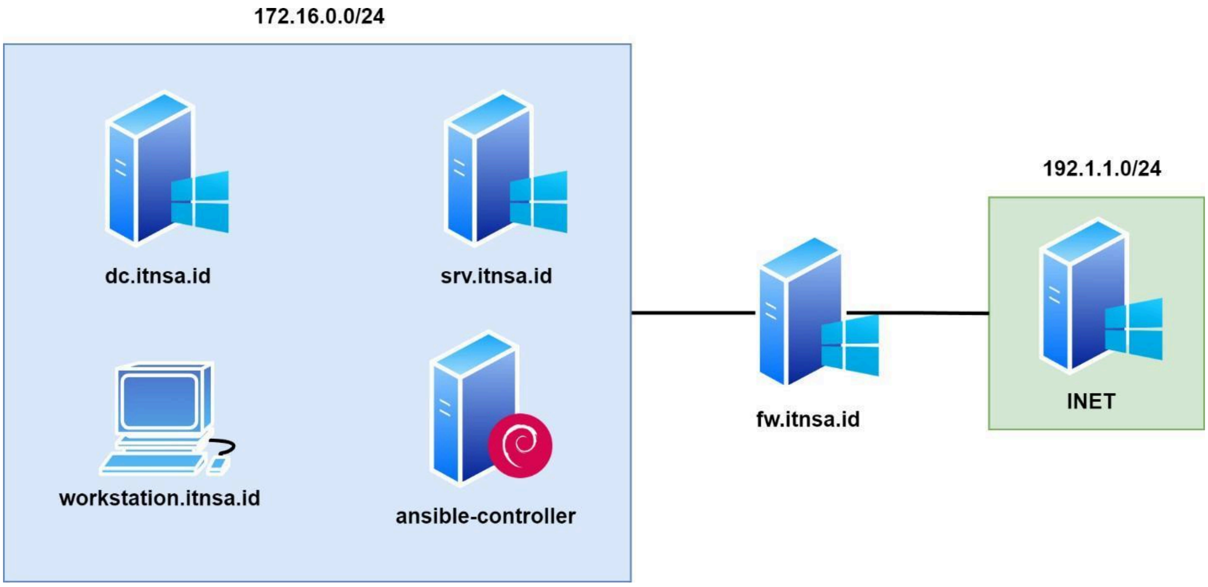
For example, creating shared folder **Test** with path **C:\Sharing\Test** while **Domain Users** will have read permission and **Manager** group have Full permission can be done using this command:

```
# ansible-playbook /etc/ansible/shared_folder.yml -e share_name=Test \
  -e path='C:/sharing/test' \
  -e access='"Domain Users"' \
  -e owner='Manager'

Playbook executed...
```

# Appendix

**Topology Diagram**




**Information Table**

| Hostname | FQDN | IP Address | Services | Domain Joined |
|---|---|---|---|---|
| dc | dc.itnsa.id | 172.16.0.1/24 | Active Directory, Certificate Authority, DNS | Yes |
| srv | srv.itnsa.id | 172.16.0.10/24 | IIS Web Service, File Server | Yes |
| fw | fw.itnsa.id | 172.16.0.254/24<br>192.1.1.1/24 | DHCP, Routing | Yes |
| workstation | workstation.itnsa.id | DHCP | - | Yes |
| ansible-srv | - | 172.16.0.151/24 | Ansible | No |

| | | | | |
|---|---|---|---|---|
| INET | - | 192.1.1.100/24 | DNS, IIS Web Service | No |